Phage Propagation Plaque Assay (Small Scale) Protocol

2026年2月28日 星期六

To amplify bacteriophage using Xanthomonas host bacteria and generate high-titer phage lysate. Used to detect phage and confirm infection.

Materials

- Log-phase Xcc culture (OD600 ≈ 0.4–0.6)
- Phage stock
- NYG soft agar (0.6–0.7%)
- NYG agar plates
- NYG liquid medium
- 30°C incubator

Procedure

1. Mix:
   - 100 µL log-phase X
   - Serially diluted phage (10–100 µL)
2. Incubate mixture 10 min at room temperature (adsorption).
   a. Adsorption is the process by which a bacteriophage attaches to the surface of a bacterial cell before injecting its genetic material.
3. Add 3 mL molten NYG soft agar (cooled to ~40°C).
4. Pour onto NYG agar plate.
5. Let solidify.
6. Incubate at 30°C overnight.

Plaques should appear as clear zones.